Flow records to a directed communication multigraph

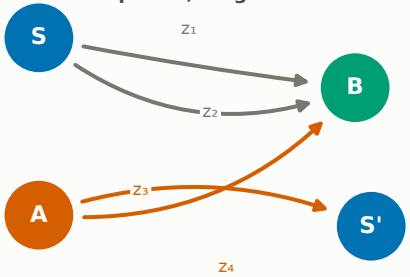
one row per flow record

ip_src	ip_dst	prt_dst	num_pkts	is_attack	edge
10.0.0.11	192.168.1.7	1883	9	0	z_1
10.0.0.11	192.168.1.7	1883	7	0	z_2
192.168.2.5	192.168.1.7	1883	4	1	z_3
192.168.2.5	10.0.0.12	43112	1	1	z_4

S = 10.0.0.11 A = 192.168.2.5 B = 192.168.1.7 S' = 10.0.0.12

rows 1 and 2 share an endpoint pair but are separate flows

nodes = IP endpoints, edges = individual flows



parallel edges are kept, so every flow keeps its own attribute vector and receives its own prediction